Impact of Election Day on Bitcoin Trading: A High-Frequency Microstructure Analysis

Final Project-FE 570

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1 The microstructure dataset

In this analysis, we use high-frequency data of XBTUSD (Bitcoin) for the day before the election, November 4, 2024, and the election day, November 5, 2024. The objective is investigation Bitcoin trading changes on election day. The number of transactions in BitMex on the day of November 4th is 72,209 and November 5th is 91496. For the intra-day analysis, the 24-hour period is divided into 8 time slots. The dataset was collected from BitMex, which provides quote and trade data separately, requiring preprocessing to merge the two datasets. The criterion for merging is the time at which the trades occurred.

2 Liquidity Measures

2.1 Quoted spread and Effective spread

We aim to conduct a study of intra-day liquidity. First, the *getLiquidityMeasures* function from the *highfrequency* package is used to estimate the quoted spread, effective spread, and realized spread.

The quoted spread (QS) is obtained simply as the difference between the bid and the ask price. The effective spread (ES) measures the immediate profit earned by a market maker (liquidity provider) who placed a limit order that was executed by a market order.

Table 1: 1 day Quoted spread and Effective spread

	Avg. QS	Avg. ES	Avg. RS
2024.11.04	6.7188	8.9604	6.4800
2024.11.05	9.6223	6.8302	0.7188

2.2 Intra-day Analysis

Unlike stocks, Bitcoin can be traded 24/7 year-round. Therefore, for intraday analysis, the 24-hour period is divided into 8 time slots, each spanning 3 hours.

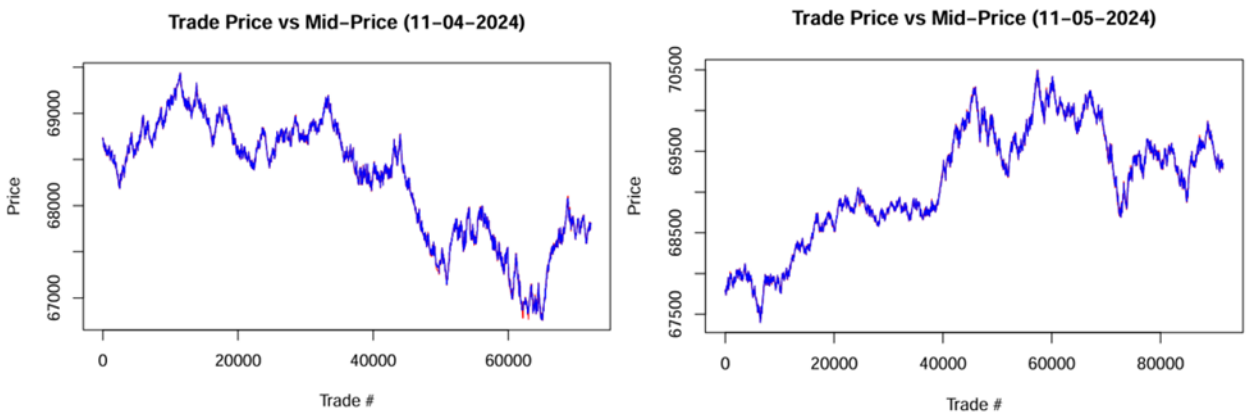
Table 2: Intra-day Quoted spread and Effective spread, 2024.11.04

	00 ~ 03	03 ~ 06	06 ~ 09	09 ~ 12	12 ~ 15	15 ~ 18	18 ~ 21	21 ~ 24
ES	0.1676	1.6492	0.9668	0.6965	1.1172	3.9993	0.8106	4.3659
QS	5.9630	5.2048	4.7835	5.5885	8.3063	11.0382	8.8591	15.5734

Table 3: Intra-day Quoted spread and Effective spread, 2024.11.05

	00 ~ 03	03 ~ 06	06 ~ 09	09 ~ 12	12 ~ 15	15 ~ 18	18 ~ 21	21 ~ 24
ES	4.9231	4.6450	3.9058	4.5158	7.4185	8.0675	9.6863	6.883
QS	7.3192	6.5875	5.6375	6.3093	10.7255	11.3310	12.8089	10.1481

Fig.1: Comparison of Trade Price vs Mid-Price

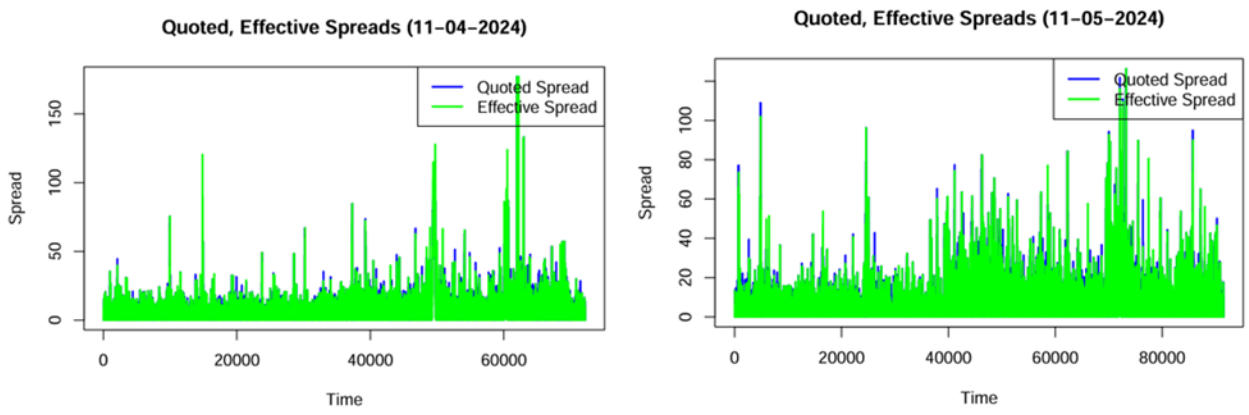


4th November 2024: Trading begins with a peak and gradually declines throughout the day. While there are some smaller fluctuations, the overall trend is downward, with a slight recovery toward the end. The price shows moderate variability, with significant declines occurring in the latter half of the day.

5th November 2024: Trading starts with a small increase, followed by steady upward momentum. Peaks occur around the middle, after which the latter part of the session sees a sideways trend. Volatility seems higher than on the previous day, marked by a wider range of prices and more significant spikes in movement.

On both days, the Mid-price (average of bid and ask prices) closely tracks the trade prices, which indicates efficient market pricing and heightened market activity.

Fig.2: Comparison of Quoted &Effective Spread

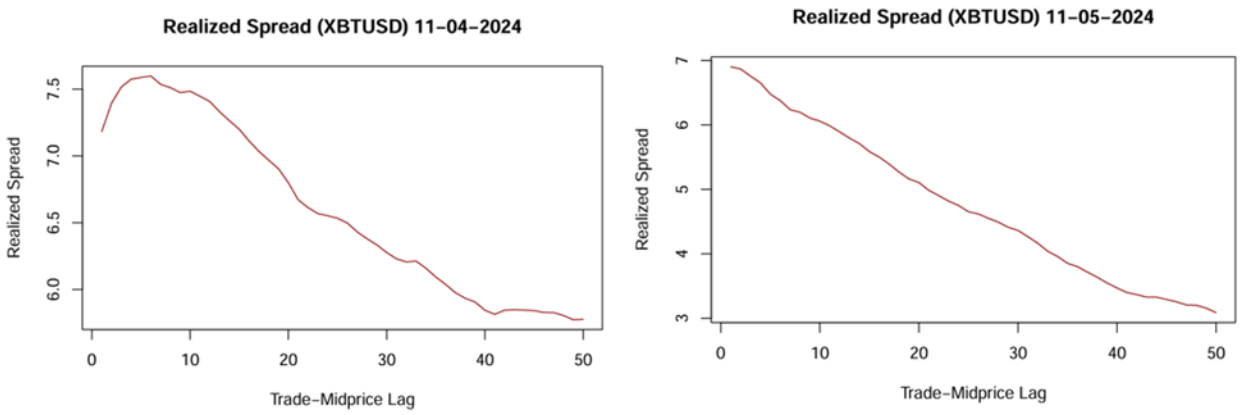


4th November 2024: Both the quoted and effective spreads display higher overall levels and greater volatility. The quoted spread is generally low, with most values clustering below 50, but has occasional spikes exceeding 100. The effective spread shows even more variability, frequently spiking above 50 and occasionally surpassing 150. This volatility, particularly in the effective spread, indicates larger fluctuations in the execution costs.

5th November 2024: In contrast, the spreads are lower but more stable, especially with the quoted spread consistently staying under 40 and the effective spread showing reduced variability staying below 100 for most of the session.

Overall, the election day exhibited compressed price spreads and reduced volatility, indicating improved market conditions or greater trading efficiency while the previous day reflected higher volatility and larger spread magnitudes, suggesting lower market liquidity or increased trading frictions.

Fig.3: Comparison of Realized Spread



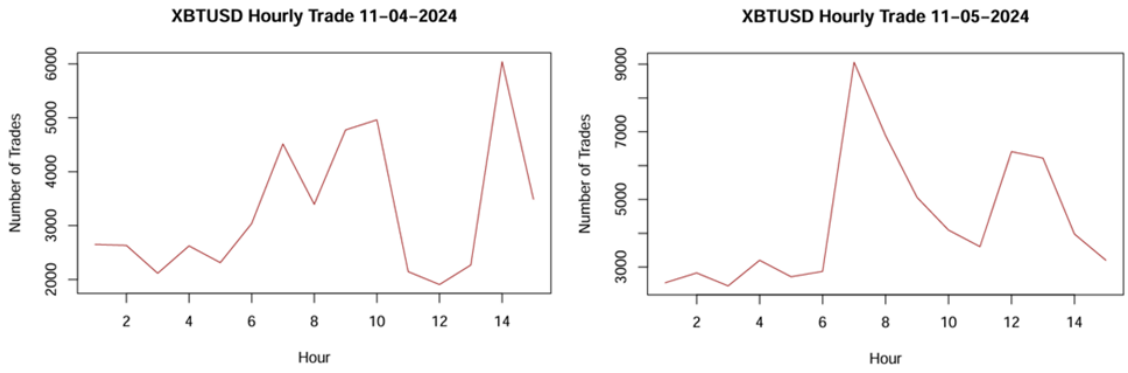
4th November 2024: The realized spread is high around 7.5 units when the trade-midprice lag is 0. Initially, the spread experiences a slight rise but then begins to decline consistently as the lag increases. There is a sharp decline around lag 10, resulting in a final spread below 6.0 at lag 50. This non-linear decline suggests higher market volatility in the early stages, with fluctuations indicating less predictability in the market behavior for that day.

5th November 2024: the realized spread begins around 7.0 units and decreases steadily in a near-linear fashion throughout the day. There is no early peak like the previous day, and the spread consistently decreases from the beginning. By the time the lag reaches 50, the realized spread has dropped to 3. This represents a sharper, more consistent decline compared to the previous day, suggesting a smoother, less volatile market with more predictable trading dynamics.

The realized spread increases as the trade-midprice lag increases indicating that market participants earn smaller spreads over extended time as the market undergoes adjustments. The high initial spread values reflect the cost of trades executed while the decreasing line shows the price efficiency. The downward trend also indicates that the market is going through price correction, which reduces the initial profit captured during early trades.

The overall change is more pronounced on the election day.

Fig.4: Comparison of Hourly Trade

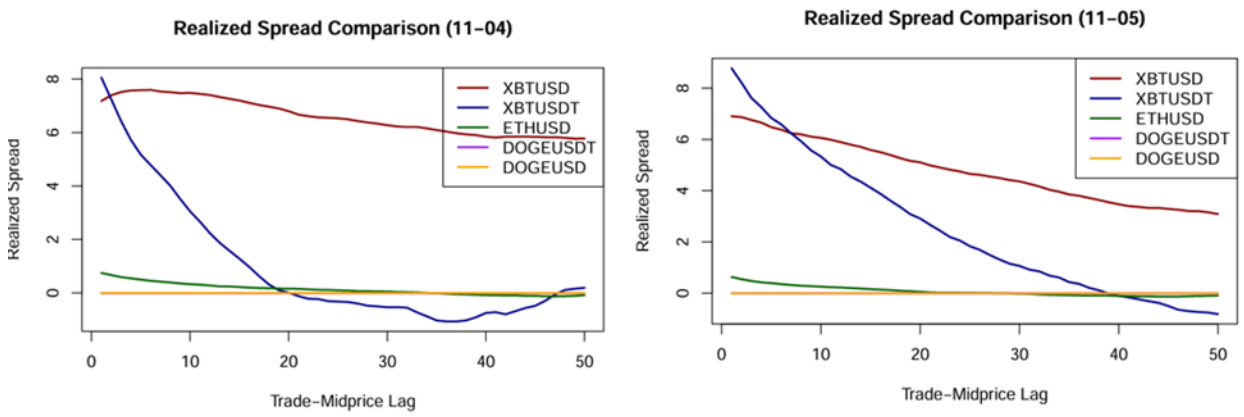


4th November 2024: The hourly trade fluctuates between 2,000 and 6,000, and significantly high around 8.00 AM and 14.00 PM, and down around 12.00 PM. This pattern shows significant variability throughout the day, with trading volumes distributed more evenly across different hours. The fluctuation in volume suggests that market activity was more spread out, with multiple trading during the day.

5th November 2024: The hourly trade volumes are generally higher, ranging between 3,000 and 9,000. There is a steep rise in trade around 13.00 PM – 18.00 PM, followed by a steady decline, with a minor increase between 19.00 PM – 22.00 PM. The pattern is more consistent, showing a clear decline in trading volume after the morning peak.

The overall high trading activity on the election day, particularly in the early hours, suggests a more active and engaged market, potentially driven by different market conditions and trading behaviors compared to the previous day. This shift could indicate increased market participation or differing investor sentiment between the two days.

Fig.5: Comparison between Top Five Trades



4th November 2024: *XBTUSD* has the highest realized spread, starting around 8 and gradually declining to around 6 by lag 50. *XBTUSDT* begins with a similar higher spread but experiences a steep decline around lag 30, dropping to near zero before rising slightly. *ETHUSD* and *DOGEUSDT* show relatively lower spreads, remaining flat throughout the day. *DOGEUSD* has the lowest and the most stable spread, staying near zero throughout all lags.

5th November 2024: *XBTUSD* shows a steady decline around 8 and dropping to around 4. *XBTUSDT* follows a similar trend, beginning high around 8 and declining sharply below zero by lag 50. *ETHUSD*, *DOGEUSDT* and *DOGEUSD* remain consistently flat, with *DOGEUSD* staying near zero. Compared to the previous day, the realized spread for *XBTUSD* shows a severe decline indicating a more significant change in the trading behavior.

In comparison, *XBTUSD* and *XBTUSDT* on both days show similar trends, with *XBTUSDT* displaying more volatility, particularly before the election day. However, the realized spread for *XBTUSD* witnessed significant decline on the election day, indicating a major shift in the market and trading behavior dynamics

3 Volatility measure

3.1 Roll model and Realized volatility

To estimate the log normal volatility, we use the Roll estimation model. When calculating annual volatility, stocks are typically adjusted multiplying by √252, as they are traded on 252 trading days in a year. However, since Bitcoin is traded 365 days a year, the adjustment is made by multiplying by √365.

Table 3: Annualized log-normal volatility of XBTUSD

	Roll model	Realized volatility
2024.11.04	0.3554	0.3569
2024.11.05	0.4095	0.4081

Table 4 shows the annualized log-normal volatility of *XBTUSD* for November 4th and 5th. The volatility of Bitcoin at 35% is not as high as it might seem. Stock ETFs typically have an annual volatility of 10% to 15%, and Bitcoin has seen volatility rise to as high as 200% in some cases. This is likely because BitMEX is not the primary exchange, and its data does not fully reflect the overall trading volume of Bitcoin.

The volatility estimated by Roll model and realized volatility values are very close on both dates, suggesting that the Roll model accurately approximates the realized volatility for this high-frequency data set. The Roll model reflects market friction costs (e.g., spread) and small price changes, and its alignment with realized volatility supports the reliability of this model. The volatility increases from November 4th to November 5th, indicating greater price variability and trading volumes in Bitcoin during the election day. Events like election days are likely to trigger aggressive trading in the crypto market. Additionally, since the Roll model demonstrates a high correlation with past realized volatility, it could be useful for predicting future volatility.

3.2 Intra-day volatility measure

Table 4: Intra-day Annualized volatility measure

	00 ~ 03	03 ~ 06	06 ~ 09	09 ~ 12	12 ~ 15	15 ~ 18	18 ~ 21	21 ~ 24
11.04	0.3370	0.2245	0.2313	0.2469	0.3743	0.4624	0.3692	0.5375
11.05	0.2492	0.1663	0.1666	0.1707	0.3817	0.4739	0.4727	0.3025

Table 5: Number of trades for each time slot

	00 ~ 03	03 ~ 06	06 ~ 09	09 ~ 12	12 ~ 15	15 ~ 18	18 ~ 21	21 ~ 24
11.04	10873	5653	6320	7249	9865	13131	6325	12793
11.05	10278	6622	7431	8470	14622	16003	16236	11834

Fig.6: Intra-day volatilities for each day

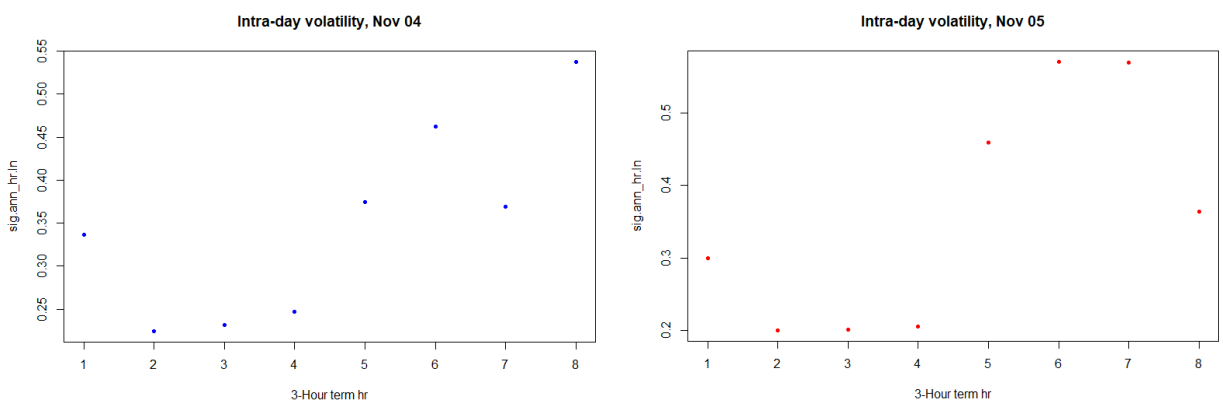


Table 5 and 6 show the volatilities and trading volumes for each timeline which is split as same as the intra-day liquidity measure in Section 2. According to Table 5, there is higher volatility in evening or nighttime especially 21:00-24:00 and 15:00-18:00. Inversely, there is lower volatility in the early morning, indicating reduced market activity during these times. The proportional relationship between trading volume and volatility is obvious, but the highest volatility from 21:00 to 24:00 on November 4th matches the second highest trading volume, which implies more drastic price changes occurred during that time.

4 PIN Model Estimation

4.1 PIN model estimation for 11/05/2024 (Election Day)

High Proportion of Informed Trades alpha, α=0.750031. A significant share of trade was driven by informed participants. This is likely due to the heightened market activity and uncertainty on election day, as informed traders capitalize on their knowledge of potential outcomes and their impact on Bitcoin prices.

The PIN value of 0.2653 (26.53%) indicates a moderate level of informed trading on November 5, 2024, coinciding with the election day. This indicates that traders with superior knowledge of Bitcoin's future price movements executed roughly 25% of all trades.

Fig.7: PIN measure for 11/05/2024

```
      Buy Sell
[1,] 4562 5716
[2,] 4210 2412
[3,] 4109 3322
[4,] 3435 5035
[5,] 7379 7243
[6,] 8224 7779
[7,] 6920 9316
[8,] 5445 6389
```

```
PIN model

==========  ===========
Variables   Estimates
==========  ===========
alpha       0.750031
delta       1
mu          4046.01
eps.b       5535.5
eps.s       2866.99
----
Likelihood  (2942.597)
PIN         0.265332
==========  ===========
```

The relatively higher rate of buy-side uninformed trades (eps.b = 5535.5) compared to sell-side uninformed trades (eps.s =2866.99) suggests bullish sentiment among uninformed traders during this period.

4.2 PIN model estimation for 11/04/2024

The trading activity on November 4, 2024, reflects a market primarily driven by routine transactions, with a moderate level of informed participation of Bitcoin trades.

Fig.8: PIN measure for 11/04/2024

```
> data
      Buy Sell
[1,] 4313 6560
[2,] 2930 2723
[3,] 3252 3068
[4,] 3607 3642
[5,] 4070 5795
[6,] 6266 6865
[7,] 3165 3160
[8,] 6497 6296
```

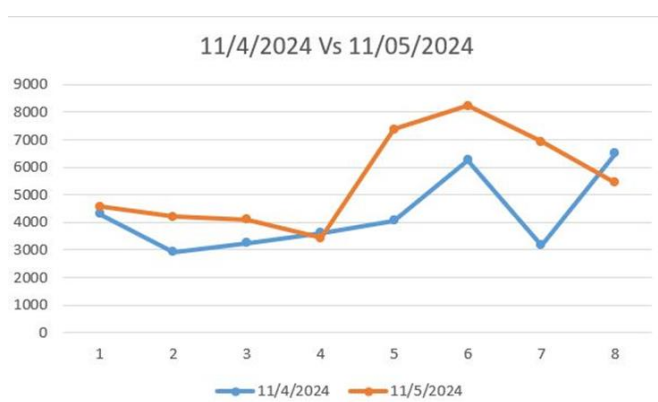
```
PIN model

==========  ===========
Variables   Estimates
==========  ===========
alpha       0.499997
delta       1
mu          3230.75
eps.b       4262.5
eps.s       3148.25
----
Likelihood  (1696.35)
PIN         0.178966
==========  ===========
```

We found Proportion of Informed Trading (α) 0.4999, indicates that while there was some strategic trading activity, it did not dominate the Bitcoin market. A PIN of 17.9% (0.1789) suggests that most market activity was regular and driven by general trading flows. The lower proportion of strategic trade reflects a market environment characterized by typical conditions without significant external drivers. We found Delta=1, which indicates all trades aligned with the actual market movements, indicating precise alignment between trade execution and prevailing market trends.

4.3 Comparison of November 4, 2024, vs. November 5, 2024 (Election Day)

The arrival rate of strategic trades (mu) was higher on November 5 (4046.01) compared to November 4 (3230.75), consistent with increased market activity.

Fig.8: Comparison between buy-side of November 4, 2024, and November 5



This graph compares the buy-side trading activity between November 4, 2024, and November 5, 2024. It highlights the increase in the PIN on November 5, 2024, indicating a higher proportion of trades driven by informed participants.

Table 7: PIN measure coefficients

Date	Alpha	Delta	mu	eps.b	eps.s	Likelihood	PIN
11/05/2024	0.750031	1	4046.01	5535.5	2866.99	2942.59	0.265332
11/04/2024	0.49997	1	3230.75	4262.5	3148.25	1696.35	0.178966

Buy-side general trade (eps.b) was more frequent than sell-side trades (eps.s) on both days, suggesting persistent bullish sentiment across the period. However, the gap between eps.s and eps.b narrowed on November 5, reflecting more balanced trading as the market reacted to election-related developments.

The higher PIN value on November 5 (26.53%) compared to November 4 (17.90%) further confirms an increase in strategic trading activity. The election catalyzed heightened market participation from traders acting on market-relevant insights.

The comparison reveals a significant shift in Bitcoin market dynamics between November 4 and November 5, 2024. While November 4 reflected routine trading activity with low strategic intensity, November 5 saw a marked increase in strategic trading as market participants reacted to election developments.

Appendix

```
library(TTR)
library(timeDate)
library(quantmod)
library(InfoTrad)
library(xts)
library(highfrequency)
library(data.table)
library(dplyr)
rm(list=ls())

## 1. Data setup
#   load data set
options(digits.secs=3)
Sys.setenv(TZ='GMT')

# Set your file path to the CSV
file_path_quote <- "C:/Users/kdw00/OneDrive - stevens.edu/Desktop/Stevens
course work/24Fall/FE570/Project/20241105_quote.csv"
file_path_trade <- "C:/Users/kdw00/OneDrive - stevens.edu/Desktop/Stevens
course work/24Fall/FE570/Project/20241105_trade.csv"

# Load the CSV data into R as a data frame
qdata <- read.csv(file_path_quote)
tdata <- read.csv(file_path_trade)
tdata <- tdata[,!names(tdata) %in% c("trdMatchID", "grossValue",
"homeNotional","foreignNotional", "trdType")]

tradesSY <- as.data.frame(table(tdata$symbol))
tradesSY_sorted <- tradesSY[order(-tradesSY$Freq),]
top5 <- head(tradesSY_sorted,5)
top5

## 2. Liquidity analysis
# 1) quoted spread, effective spread, realized spread
# Filter for XBTUSD trades
qdata$timestamp <- gsub("D", " ", qdata$timestamp)
qdata$timestamp <- as.POSIXct(qdata$timestamp, format = "%Y-%m-%d %H:%M:%OS",
tz="EST")
tdata$timestamp <- gsub("D", " " , tdata$timestamp)
tdata$timestamp <- as.POSIXct(tdata$timestamp, format = "%Y-%m-%d %H:%M:%OS",
tz="EST")

colnames(tdata)[colnames(tdata) == "timestamp"] <- "DT"
colnames(qdata)[colnames(qdata) == "timestamp"] <- "DT"

tdata$DT <- as.POSIXct(tdata$DT)
qdata$DT <- as.POSIXct(qdata$DT)
```

```r
## Checking for NA values
sum(is.na(qdata$askprice))
sum(is.na(qdata$bidprice))
sum(is.na(tdata$price))


colnames(qdata)[colnames(qdata) == "bidSize"] <- "BIDSIZE"
colnames(qdata)[colnames(qdata) == "bidPrice"] <- "BID"
colnames(qdata)[colnames(qdata) == "askSize"] <- "OFRSIZE"
colnames(qdata)[colnames(qdata) == "askPrice"] <- "OFR"

head(qdata)

## Converting dataframe to data table
qdata <- as.data.table(qdata)
tdata <- as.data.table(tdata)
tqdata <- matchTradesQuotes(tdata,qdata)

## Plotting the price for XBTUSD trades
tqdata.test <- tqdata[tqdata$SYMBOL == 'XBTUSD']

head(tqdata.test)
tail(tqdata.test)

length(tqdata.test$SIZE)

asks <- as.numeric(tqdata.test$OFR)
bids <- as.numeric(tqdata.test$BID)
mids <- 0.5*bids + 0.5*asks

pmin <- min(as.numeric(tqdata.test$PRICE))
pmax <- max(as.numeric(tqdata.test$PRICE))

plot(as.numeric(tqdata.test$PRICE), col="red", type="l", ylab="Trade Price",
     xlab="Trade #", main="Trade Price (Intraday)", ylim=c(pmin-0.1,
pmax+0.1))
lines(mids, type="l", col="blue")


plot(as.numeric(tqdata.test$SIZE), col="red", type="l", xlab="Trade #",
     ylab="Trade Size", main="Trade Volume", ylim=c(0,100000))


## liquidity measures for XBTUSD
liq.ms <- getLiquidityMeasures(tqdata.test)

q.s <- (liq.ms$quotedSpread)
e.s <- (liq.ms$effectiveSpread)
```

```r
# Intra-day liquidity, by exchange 3 hours
# Define time ranges as POSIXct
hr1_start <- as.POSIXct("2024-11-05 00:00:00", tz = "EST")
hr1_end <- as.POSIXct("2024-11-05 03:00:00", tz = "EST")
hr2_start <- as.POSIXct("2024-11-05 03:00:00", tz = "EST")
hr2_end <- as.POSIXct("2024-11-05 06:00:00", tz = "EST")
hr3_start <- as.POSIXct("2024-11-05 06:00:00", tz = "EST")
hr3_end <- as.POSIXct("2024-11-05 09:00:00", tz = "EST")
hr4_start <- as.POSIXct("2024-11-05 09:00:00", tz = "EST")
hr4_end <- as.POSIXct("2024-11-05 12:00:00", tz = "EST")
hr5_start <- as.POSIXct("2024-11-05 12:00:00", tz = "EST")
hr5_end <- as.POSIXct("2024-11-05 15:00:00", tz = "EST")
hr6_start <- as.POSIXct("2024-11-05 15:00:00", tz = "EST")
hr6_end <- as.POSIXct("2024-11-05 18:00:00", tz = "EST")
hr7_start <- as.POSIXct("2024-11-05 18:00:00", tz = "EST")
hr7_end <- as.POSIXct("2024-11-05 21:00:00", tz = "EST")
hr8_start <- as.POSIXct("2024-11-05 21:00:00", tz = "EST")
hr8_end <- as.POSIXct("2024-11-05 24:00:00", tz = "EST")


# Filter data for the time range
tqdata.1 <- tqdata.test[DT >= hr1_start & DT < hr1_end]
tqdata.2 <- tqdata.test[DT >= hr2_start & DT < hr2_end]
tqdata.3 <- tqdata.test[DT >= hr3_start & DT < hr3_end]
tqdata.4 <- tqdata.test[DT >= hr4_start & DT < hr4_end]
tqdata.5 <- tqdata.test[DT >= hr5_start & DT < hr5_end]
tqdata.6 <- tqdata.test[DT >= hr6_start & DT < hr6_end]
tqdata.7 <- tqdata.test[DT >= hr7_start & DT < hr7_end]
tqdata.8 <- tqdata.test[DT >= hr8_start & DT < hr8_end]

liq.1 <- getLiquidityMeasures(tqdata.1)
liq.2 <- getLiquidityMeasures(tqdata.2)
liq.3 <- getLiquidityMeasures(tqdata.3)
liq.4 <- getLiquidityMeasures(tqdata.4)
liq.5 <- getLiquidityMeasures(tqdata.5)
liq.6 <- getLiquidityMeasures(tqdata.6)
liq.7 <- getLiquidityMeasures(tqdata.7)
liq.8 <- getLiquidityMeasures(tqdata.8)

es.hr <- c()
es.hr[1] <- abs(mean(as.numeric(liq.1$effectiveSpread)))
es.hr[2] <- abs(mean(as.numeric(liq.2$effectiveSpread)))
es.hr[3] <- abs(mean(as.numeric(liq.3$effectiveSpread)))
es.hr[4] <- abs(mean(as.numeric(liq.4$effectiveSpread)))
es.hr[5] <- abs(mean(as.numeric(liq.5$effectiveSpread)))
es.hr[6] <- abs(mean(as.numeric(liq.6$effectiveSpread)))
es.hr[7] <- abs(mean(as.numeric(liq.7$effectiveSpread)))
es.hr[8] <- abs(mean(as.numeric(liq.8$effectiveSpread)))

qs.hr <- c()
```

```r
qs.hr[1] <- mean(as.numeric(liq.1$quotedSpread))
qs.hr[2] <- mean(as.numeric(liq.2$quotedSpread))
qs.hr[3] <- mean(as.numeric(liq.3$quotedSpread))
qs.hr[4] <- mean(as.numeric(liq.4$quotedSpread))
qs.hr[5] <- mean(as.numeric(liq.5$quotedSpread))
qs.hr[6] <- mean(as.numeric(liq.6$quotedSpread))
qs.hr[7] <- mean(as.numeric(liq.7$quotedSpread))
qs.hr[8] <- mean(as.numeric(liq.8$quotedSpread))

plot(1:8,es.hr, type="p", pch=20, col="blue",
     main="EffectiveSpread", xlab="3-Hour term")

# Plot the results
# Liquidity measures in highfrequency

## Volatility from Roll model
p <- as.numeric(tqdata.test$PRICE)
dp <- diff(p)

covdp <- acf(dp, lag.max=10,
             type="covariance", plot=TRUE,
             main="Autocovariance of price changes")

gamma0 <- covdp$acf[1]
gamma1 <- covdp$acf[2]
sig2u = gamma0 + 2*gamma1
sigu <- sqrt(sig2u)

n.trades <- dim(tqdata.test)[1]
av.price <- mean(p)
av.price

#daily volatility
sig.day <- sqrt(sig2u*n.trades)
sig.day

#annualized volatility
sig.ann <- sqrt(365)*sig.day
sig.ann

# log-normal volatility
sig.ann.ln <- sig.ann/av.price
sig.ann.ln     #30.55%

# compare with the total volatility
sig.day.total <- sqrt(gamma0*n.trades)
sig.day.total

sig.ann.total <- sqrt(365)*sig.day.total
sig.ann.total
```

```r
sig.ann.ln.total <- sig.ann.total/av.price
sig.ann.ln.total

## Volatiliy realized
log_dp <- diff(log(tqdata.test$PRICE))
var_log_dp <- var(log_dp)
vol_log_dp <- sqrt(var_log_dp)
ann_vol_log_dp <- vol_log_dp * sqrt(365 * n.trades)
ann_vol_log_dp

## Get signature plot
p <- as.numeric(tqdata.test$PRICE)
realizedVar <- function(q){rCov(diff(p, lag=q, differences=1))/q}
realizedVar(1)
sqrt(realizedVar(1))

# compute the signature plot RV(lag)

rv_data <- NULL

for(q in 1:300){
  rv_data <- c(rv_data, realizedVar(q))
}

plot(rv_data, type ="l", main="Signature plot")
q5min <- n.trades*5/1440
rv5 = realizedVar(q5min)
sqrt(rv5)
plot(rv_data, type ="l", main="Signature plot for prices",
     xlab="lag")
abline(h=rv5,col="red")

rv1 = realizedVar(1)
sqrt(365)*sqrt(rv1)/av.price

## Intra-day volatility
p1 <- as.numeric(tqdata.1$PRICE)
p2 <- as.numeric(tqdata.2$PRICE)
p3 <- as.numeric(tqdata.3$PRICE)
p4 <- as.numeric(tqdata.4$PRICE)
p5 <- as.numeric(tqdata.5$PRICE)
p6 <- as.numeric(tqdata.6$PRICE)
p7 <- as.numeric(tqdata.7$PRICE)
p8 <- as.numeric(tqdata.8$PRICE)

dp1 <- diff(p1)
dp2 <- diff(p2)
dp3 <- diff(p3)
dp4 <- diff(p4)
```

```r
dp5 <- diff(p5)
dp6 <- diff(p6)
dp7 <- diff(p7)
dp8 <- diff(p8)

covdp1 <- acf(dp1, lag.max=10, type="covariance", plot=FALSE)
covdp2 <- acf(dp2, lag.max=10, type="covariance", plot=FALSE)
covdp3 <- acf(dp3, lag.max=10, type="covariance", plot=FALSE)
covdp4 <- acf(dp4, lag.max=10, type="covariance", plot=FALSE)
covdp5 <- acf(dp5, lag.max=10, type="covariance", plot=FALSE)
covdp6 <- acf(dp6, lag.max=10, type="covariance", plot=FALSE)
covdp7 <- acf(dp7, lag.max=10, type="covariance", plot=FALSE)
covdp8 <- acf(dp8, lag.max=10, type="covariance", plot=FALSE)

sigu_hr <- c()
sig2u_1 = covdp1$acf[1] + 2*covdp1$acf[2]
sigu_hr[1] <- sqrt(sig2u_1)
sig2u_2 = covdp2$acf[1] + 2*covdp2$acf[2]
sigu_hr[2] <- sqrt(sig2u_2)
sig2u_3 = covdp3$acf[1] + 2*covdp3$acf[2]
sigu_hr[3] <- sqrt(sig2u_3)
sig2u_4 = covdp4$acf[1] + 2*covdp4$acf[2]
sigu_hr[4] <- sqrt(sig2u_4)
sig2u_5 = covdp5$acf[1] + 2*covdp5$acf[2]
sigu_hr[5] <- sqrt(sig2u_5)
sig2u_6 = covdp6$acf[1] + 2*covdp6$acf[2]
sigu_hr[6] <- sqrt(sig2u_6)
sig2u_7 = covdp7$acf[1] + 2*covdp7$acf[2]
sigu_hr[7] <- sqrt(sig2u_7)
sig2u_8 = covdp8$acf[1] + 2*covdp8$acf[2]
sigu_hr[8] <- sqrt(sig2u_8)

n.trades_hr <- c()
n.trades_hr[1] <- dim(tqdata.1)[1]
n.trades_hr[2] <- dim(tqdata.2)[1]
n.trades_hr[3] <- dim(tqdata.3)[1]
n.trades_hr[4] <- dim(tqdata.4)[1]
n.trades_hr[5] <- dim(tqdata.5)[1]
n.trades_hr[6] <- dim(tqdata.6)[1]
n.trades_hr[7] <- dim(tqdata.7)[1]
n.trades_hr[8] <- dim(tqdata.8)[1]

av.price_hr <- c()
av.price_hr[1] <- mean(p1)
av.price_hr[2] <- mean(p2)
av.price_hr[3] <- mean(p3)
av.price_hr[4] <- mean(p4)
av.price_hr[5] <- mean(p5)
av.price_hr[6] <- mean(p6)
av.price_hr[7] <- mean(p7)
```

```r
av.price_hr[8] <- mean(p8)

sig.3hr <- c()
sig.3hr <- sqrt(n.trades_hr) * sigu_hr
sig.ann_hr <- sqrt(365*8)*sig.3hr
sig.ann_hr.ln <- sig.ann_hr/av.price_hr

plot(1:8, sig.ann_hr.ln, type="p", pch=20, col="red",
     main="Intra-day volatility, Nov 05", xlab="3-Hour term hr")

plot(1:8, n.trades_hr, type="p", pch=20, col="red",
     main="Trade volumes, Nov 05", xlab="3-Hour term hr")


### PIN measure
x.1 <- getTradeDirection(tqdata.1)
x.2 <- getTradeDirection(tqdata.2)
x.3 <- getTradeDirection(tqdata.3)
x.4 <- getTradeDirection(tqdata.4)
x.5 <- getTradeDirection(tqdata.5)
x.6 <- getTradeDirection(tqdata.6)
x.7 <- getTradeDirection(tqdata.7)
x.8 <- getTradeDirection(tqdata.8)

td.1 <- matrix(x.1)
td.2 <- matrix(x.2)
td.3 <- matrix(x.3)
td.4 <- matrix(x.4)
td.5 <- matrix(x.5)
td.6 <- matrix(x.6)
td.7 <- matrix(x.7)
td.8 <- matrix(x.8)

buy.1 <- which(td.1 > 0)
n.buy.1 <- length(matrix(buy.1))
n.sell.1 <- length(td.1) - n.buy.1

buy.2 <- which(td.2 > 0)
n.buy.2 <- length(matrix(buy.2))
n.sell.2 <- length(td.2) - n.buy.2

buy.3 <- which(td.3 > 0)
n.buy.3 <- length(matrix(buy.3))
n.sell.3 <- length(td.3) - n.buy.3

buy.4 <- which(td.4 > 0)
n.buy.4 <- length(matrix(buy.4))
n.sell.4 <- length(td.4) - n.buy.4

buy.5 <- which(td.5 > 0)
```

```r
n.buy.5 <- length(matrix(buy.5))
n.sell.5 <- length(td.5) - n.buy.5

buy.6 <- which(td.6 > 0)
n.buy.6 <- length(matrix(buy.6))
n.sell.6 <- length(td.6) - n.buy.6

buy.7 <- which(td.7 > 0)
n.buy.7 <- length(matrix(buy.7))
n.sell.7 <- length(td.7) - n.buy.7

buy.8 <- which(td.8 > 0)
n.buy.8 <- length(matrix(buy.8))
n.sell.8 <- length(td.8) - n.buy.8


Buy <- c(n.buy.1, n.buy.2, n.buy.3, n.buy.4, n.buy.5, n.buy.6, n.buy.7,
n.buy.8)
Sell <- c(n.sell.1, n.sell.2, n.sell.3, n.sell.4, n.sell.5, n.sell.6,
n.sell.7, n.sell.8)
data <- cbind(Buy, Sell)

library(PINstimation)

estimate <- pin_ea(data) ## PIN model estimation
estimate_adjpin <- adjpin(data, initialsets = "GE") ## adjust PIN model
estimate
estimate_adjpin
```